Composite Perception and Multiscale Fusion Network for Arbitrary-Oriented Object Detection in Remote Sensing Imagery

Peng Bai, Ying Xia, *Member, IEEE*, and Jiangfan Feng

***Abstract*— Arbitrary-oriented object detection has been widely used in remote sensing image object detection, mainly benefiting from accurate boundary boxes with less background information. However, the fixed receptive field feature extraction adopted by current mainstream methods limits the dynamic focusing ability of the model, resulting in insufficient context information capture, which leads to small object submergence. Furthermore, the complex background and dense distribution of objects in remote sensing images also increase the difficulty of detection, making it vulnerable to noise and fuzzy boundary interference. Therefore, we propose a new arbitrary-oriented object detection (AOOD) network called composite perception and multiscale fusion network (CPMFNet). First, to aggregate different levels of context information of the object and effectively enhance the context modeling ability of the detector, we propose an adaptive spatial perception module (ASPM) and insert it into the residual network as a plug-and-play module to realize dynamic receptive field convolution. Second, we redesigned a feature pyramid network (FPN) equipped with a composite perception and context integration module (CPCIM) and detail noise balance filter (DNBF). CPCIM realizes cross-scale feature fusion, mines position and detail features of shallow networks, enhances global semantic expression, and weakens interference of background information. DNBF can effectively suppress the interference of noise and fuzzy boundaries and guide deep networks to alleviate the inundation of small objects. Experiments on four remote sensing benchmark datasets (DOTA-v1.0, DOTA-v1.5, DIOR-R, and HRSC2016) show that CPMFNet (ours) achieves SOTA performance, in which the mean average precision (mAP) is 78.23% on the DOTA-v1.0 dataset, 70.85% on the DOTA-v1.5 dataset, and 67.58% and 98.61% on the DIOR-R and HRSC2016 datasets, respectively.**

***Index Terms*— Arbitrary-oriented object detection (AOOD), feature fusion, multiscale objects, remote sensing image.**

I. Introduction

REMOTE sensing image object detection is one of the challenging tasks in the field of remote sensing. It focuses on accurately identifying each object category in the image and accurately locating its position. It has a wide range of applications in agriculture, military investigation, disaster relief, environmental monitoring, and other fields. With the regular launch of more and more professional remote sensing satellites, we can collect more remote sensing optical images. The rich information contained in them is crucial for object detection. However, remote sensing images are mostly taken from the perspective of the aerial view, with large differences in object scales, dense distribution, and diverse directions. Compared with natural images, the uniqueness of its imaging makes object detection still face some challenges and is also a research hotspot in the field of remote sensing.

In recent years, since oriented object detection can generate more accurate bounding boxes in scenes with dense distribution and disordered directions of objects, object detection using the oriented bounding box (OBB) instead of horizontal bounding box has become a mainstream trend in research and has made considerable progress in the field of remote sensing object detection. Remote sensing image arbitrary-oriented object detection (AOOD) methods can be divided into anchor-based and anchor-free detection. The anchor-based oriented object detection includes one- and two-stage detection methods. In one-stage anchor-based methods, the difficulty of feature alignment is the bottleneck to improving detection performance. R^3Det [1] used feature interpolation to reconstruct the refined bounding box position feature maps to achieve feature alignment. AO2-DETR [2] proposed a new oriented proposal generation mechanism to improve position priori of pooled features and then modulated cross-attention in the Transformer decoder. At the same time, the proposed adaptive-oriented refinement module was introduced to achieve alignment between regional features and objects. To provide more accurate object position and orientation information, especially for objects with large aspect ratios, YOLOv8-OBB [3], [4] introduced an angle parameter and utilized Gaussian distribution to represent the position and scale of objects. It employs Prob_IoU to measure the matching degree between detection boxes and ground-truth boxes, thereby enhancing the detection performance of the model in scenarios involving occlusion, changes in viewpoint, and other challenging conditions. Although the detection speed of two-stage anchor-based methods is lower than that of one-stage methods, its detection performance is significantly improved. For example, Ding et al. [5] alleviated the

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alignment problem between the horizontal bounding box and the object by proposing a supervised rotated RoI learner, learning transformation parameters from the horizontal bounding box to OBB under the annotation supervision of OBB. Oriented R-CNN [6] proposed a center point offset method to significantly reduce the parameters required to generate OBB. SASM [7] proposed a shape adaptive selection strategy to make the model more focused on the aspect ratio and dynamically calculate the optimal intersection ratio threshold of objects with different aspect ratios. To make the model pay more attention to high-quality samples, the shape adaptive measurement strategy was introduced to measure the quality of positive samples, optimize model training, and improve the performance of detecting large-aspect-ratio objects. The premise for anchor-based methods to accurately detect objects is to present a large number of anchor boxes, and the proportion of positive samples containing object information is small. To solve the problem of imbalance between positive and negative samples, some scholars have also proposed anchor-free object detection methods. For example, Li et al. [8] introduced a set of points to represent object position and direction as reference feature points to estimate object direction, which fully captures the direction features of the object. To detect objects in any direction and get rid of the restriction of rotation angle, G-Rep [9] converted object representation to Gaussian distribution and used the maximum likelihood estimation algorithm to optimize its parameters.

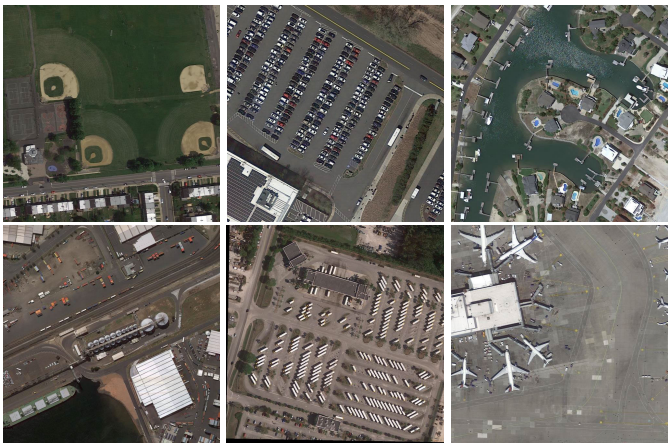
In addition, spectral images can capture a vast array of spectral information, thereby achieving highly accurate analysis and recognition of objects and scenes beyond what is possible with RGB data alone. Some scholars have also done a lot of research on this and made good progress. Li et al. [10] proposed a CasFormer especially designed for fusion-aware CHI by using dual-camera devices. It proposed spectral-aware self-attention (MSSA) to better capture spectral sequence characteristics by calculating the similarity between spectral bands and designed spatial-fused cross-attention (SFCA) to enhance spatial detail representation by embedding compact high-resolution RGB images behind the learned spectral representation. Hong et al. [11] applied a generative pretrained Transformer (GPT) model to the field of spectral remote sensing for the first time and built a foundation model that can understand and generate spectral remote sensing data. By combining GPT and spectral analysis technology, it achieved efficient processing and analysis of remote sensing data. Li et al. [12] proposed a new interpretable deep unfolding network LRR-Net, which established the connection between methods based on the low-rank model and data-driven DL techniques. The threshold and weight parameters are automatically learned in an end-to-end unsupervised manner, eliminating the need for manual parameter tuning.

To improve the accuracy and robustness of the model, some scholars have also carried out a series of optimizations on the loss function. For example, Qian et al. [13] proposed a modulation loss especially used for oriented object detection, which eliminated the error caused by inconsistent dimensions of anchor parameters, and further solved the problem of inconsistent parameter regression combined with eight-parameter regression. Chen et al. [14] proposed a new loss function pixel IoU (PIoU), which used the angle and IoU to regress the rotation bounding box, and effectively improved the detection effect of tilted objects in the image. At present, most mainstream-oriented object detection methods based on regression have the problem of boundary discontinuity. To solve this problem, Yang et al. [15] used regression loss approximation based on Gaussian–Wasserstein distance to calculate the nondifferentiable rotating IoU loss and achieved good results on multiple public datasets. Qian et al. [16] proposed a new BBR loss function smooth GIoU, which allows dynamic adjustment of gradient value, that is, the learning intensity, in different ranges of GIoU value to solve the problem of low accuracy in object localization.

In recent years, the design of backbone networks for oriented object detection has also attracted the attention of many scholars, and the research on backbone networks for feature extraction has also made significant progress. For example, Wu et al. [17] proposed that incomplete feature representation still cannot meet the requirements of effective and efficient processing of image deformation. Therefore, they proposed a new optical remote sensing image detector (ORSIm detector) that integrates multiple channel feature extraction, feature learning, fast image pyramid matching, and enhancement strategies. The proposed SFCF jointly considers the rotation-invariant channel features constructed in the frequency domain and original spatial channel features (such as color channels and gradient magnitude) and is continuously improved through a learning-based strategy to obtain high-level or semantically meaningful features. Considering the importance of background prior knowledge, LSKNet [18] introduced a large kernel selective mechanism into the backbone network, an adaptively adjusted receptive field in the feature extraction stage, and dynamically captured rich details and context information of different scale objects, achieving excellent detection performance in remote sensing object detection. To find more effective receptive field combinations than manual design, Gao et al. [19] first get rough receptive field combinations through a global to local search scheme and then proposed the expectation-guided local search scheme to further search for fine-grained receptive field combinations in dense search space. Pu et al. [20] also designed the backbone network feature extractor, different from LSKNet, and proposed an efficient adaptive rotational convolution (ARC) module to further process-oriented object detection. As a plug-and-play module, ARC can be inserted into any backbone network with a convolution layer to enhance the feature representation of rotating objects.

However, despite these advances, remote sensing object detection still faces several challenges and lacks further research. First, the object scales in remote sensing images differ greatly, when processing objects with different scales in remote sensing images, the feature representation often strongly depends on dynamic context information. As shown in Fig. 1(a), there is a significant difference in scale between the baseball diamond, tennis court, and the small vehicle in the image. Capturing rich context information of objects with different scales can help the model better understand the spatial

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(a) (b) (c)

Fig. 1. Difficulties in detecting RSI datasets. (a) There is a significant difference in scale between different objects in the image. (b) Fuzzy boundary interference exists in scenes with a dense distribution of small vehicles and large vehicles. (c) Planes and harbors display diverse rotation angles in remote sensing images.

distribution and scale changes of objects. Therefore, it is a key research challenge to dynamically adjust the receptive field of a convolutional network according to different scale objects to extract more suitable context information for the current object. Second, remote sensing images are mostly taken from an aerial view, with a wide viewing angle. In remote sensing images with high resolution, only a few pixels may belong to the detected object, resulting in an extreme imbalance between foreground information and background information. Alleviating the problem of small object inundation in such a complex background also brings great challenges to the design of oriented object detectors. Furthermore, as shown in Fig. 1(b), there are serious occlusion, overlap, and other phenomena in the scene with densely distributed objects, which will affect the ability of the model to learn high-quality foreground information, and ultimately lead to inaccurate object localization or classification errors. To meet the above challenges, many scholars have devoted themselves to improving backbone networks in recent years and most of them have considered dynamically capturing object context information, which reduces background interference to a certain extent and improves the overall detection accuracy. However, on this basis, there is still a lack of integrated improvement on the problem of small object inundation and poor model learning ability in complex backgrounds. If these three challenges are comprehensively considered, not only improving the backbone network but also making innovative designs for the problem of small object inundation and poor detection effect in complex scenes with overlapping occlusions, will the detection effect be better?

To cope with the above challenges, in this article, we propose a two-stage oriented object detection network called composite perception and multiscale fusion network (CPMFNet), which not only focuses on dynamically capturing the rich context information of different scale objects but also, on this basis, a redesigned feature pyramid network (FPN) achieves more efficient multiscale fusion and alleviates the problem of small object inundation and poor detection performance in complex scenes such as occlusion and overlap. First, the adaptive spatial perception (ASPM) module is proposed to dynamically perceive the context information of the target object to improve the context modeling ability of the detector and then the complex perception and context integration module (CPCIM) is inserted into FPN [21] to mine details and position features of the shallow network, integrate global semantic features, and weaken interference of complex background. We design a detail noise balance filter (DNBF) to suppress interference of noise and fuzzy boundary and alleviate the problem of small object inundation. We conducted a large number of experiments on DOTA-v1.0, DOTA-v1.5, DIOR-R, and HRSC2016, four public remote sensing datasets, and finally, our model also achieved SOTA performance.

To sum up, our main contributions can be summarized as follows.

1) The ASPM is proposed to mine context information of target objects with different scales and dynamically adjust the convolution receptive field size of an object through different receptive field branches to achieve adaptive spatial sensing.
2) We redesigned the FPN and introduced a large kernel selective mechanism into it. At the same time, the proposed CPCIM and DNBF are inserted to fuse global semantic features from the deep network and details, position information of the shallow network, suppress local noise, and reduce the phenomenon of small object missing detection and false detection.
3) For complex background and dense object scenes, a new oriented object detection network CPMFNet is proposed, which can effectively suppress interference of complex backgrounds, reduce boundary ambiguity, and improve the performance of oriented object detection.

II. Related Work

A. Oriented Object Detection Methods for Remote-Sensing Images

Oriented object detection in remote sensing images has always been one of the hotspots of computer vision research. With the continuous development of deep learning, more and more oriented object detection methods have been proposed, and the detection performance has been significantly improved. Oriented object detection attempts to extend general horizontal detection to a more fine-grained problem [20] by introducing OBB, which can provide more accurate position information of detected objects in densely distributed and complex background scenes. For this reason, many researchers are committed to the establishment of a specially oriented object detector.

In one-stage methods, the parameters are directly regressed, the reasoning speed is more emphasized, and the detection efficiency is improved according to dense anchor points in classification. Liu et al. [22] proposed that key points could not fully obtain feature representation of rotating objects and designed a center boundary dual-attention module (CBDA), which took advantage of different regional attention, and simultaneously used center point area mask and boundary area

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mask to capture important information of the object, it can make the model to focus more on important areas of the object. Liu et al. [23] designed a module to refine features of oriented objects on the basis of YOLOX [24]. By combining the rotated bounding box module (RBBM) and refined feature layers (RFLs) to extract features of oriented objects and align OBBs, the detection accuracy was improved, and the characteristics of fast detection were inherited.

The two-stage methods use RPN to generate rotated RoI, then use RRoI pooling to extract features of obtained rotated RoI, and finally predict OBB and separation through detection heads. Ding et al. [5] proposed an RoI Transformer to learn rotated RoI from horizontal RoI generated by RPN, which can convert horizontal bounding box into OBB with less calculation. This conversion method has achieved good results in the field of oriented object detection, but due to ignoring the problems of angle periodicity and boundary, the detection performance is poor on objects with large aspect ratios. Xu et al. [25] proposed a new representation method of OBB, which is called sliding vertex. The OBB is obtained by learning sliding offsets of four vertices on the regression branch of Fast R-CNN [26] head. Compared with the way of introducing angle, it greatly reduces error and avoids boundary problems. Xie et al. [6] proposed an effective and simple oriented object detector called oriented R-CNN, which uses the midpoint offset method to generate OBB, offsetting the midpoint of the upper edge and right side of the horizontal bounding box, respectively. This method greatly reduces the parameters required to generate OBB and inspires subsequent scholars to design OBB. Pu et al. [20] proposed an adaptive rotation convolution module to extract high-quality features of target objects in any direction. By introducing an efficient conditional computing mechanism to adapt to some objects with large direction changes in the image, they effectively improved the ability of feature expression in various backbone networks. Combined with oriented R-CNN, they greatly improved the detection effect of rotating objects. Li et al. [18] explored large kernel selective mechanisms in the field of remote sensing object detection for the first time. By designing a convolution module that can dynamically adjust the receptive field to build a backbone network LSKNet, it can more effectively deal with the feature extraction ability of different objects in complex backgrounds and can dynamically obtain rich context information for different objects. LSKNet was also combined with oriented R-CNN and finally achieved SOTA performance on the DOTA-v1.0 dataset.

B. Feature Fusion Network

After feature extraction of the input image by the backbone network, it is input into the neck network for cross-scale feature fusion. Considering the dense distribution of objects in remote sensing images, large-scale differences, and extremely complex background information outside objects, more and more feature fusion methods are gradually emerging. The methods that combine global semantic features extracted by deep networks with details, position, and other information extracted by shallow networks to generate new feature maps have shown excellent performance in object detection in remote sensing images. For example, FPN [21] has changed the way that most of the original object detection methods use top-level features for prediction, proposed a top-down way to process feature maps, and introduced the way of horizontal connection to integrate shallow and deep networks. Liu et al. [27] further improved FPN by transferring accurate positioning information of the shallow network to the deep network through the bottom-up path, thus greatly shortening the path between the deep network and shallow network. Dong et al. [28] integrated a multilevel feature fusion module using attention into FPN to improve the detection accuracy of rotating objects. Liu et al. [29] proposed ABNet to solve problems of large-scale difference, sparse, and unbalanced distribution of remote sensing image objects. To reduce the loss of semantic information, Zhen et al. [30] proposed a two-stage detector for adaptive feature fusion. A multiscale feature fusion module was established in FPN and combined with a weighted RoI feature aggregation module to effectively fuse features and reduce loss of background and semantic information. Lin et al. [31] improved FPN structure and introduced focal loss to alleviate serious imbalance between positive and negative samples in a one-stage network, the proposed RetinaNet realized that a one-stage network surpassed a two-stage network for the first time. Wu et al. [32] embedded a tiny U-Net into a larger U-Net backbone network and proposed an interactive cross-attention nested U-Net network called UIUNet, which achieves multilevel and multiscale representation learning of objects. The proposed RMDS integrates the residual U-block into a deep supervision network to generate deep multiscale resolution features while learning global context information, further improving the local context contrast and avoiding information loss of small objects during downsampling. In addition, to capture long-range dependencies between pixel-based objects, the IC-A module is proposed to replace the skip layer in U-Net by interactively cross-encoding low-level details and high-level semantic features.

C. Attention Mechanism

Attention mechanisms in image processing can realize plug and play, which has become one of the most popular and important technologies in the field of deep learning. SENet [33] proposed a channel attention mechanism for the first time. It learned feature weights of different channels through loss function, explicitly modeled interdependence between feature channels, and dynamically adjusted original data according to importance. ECA [34] improved an incentive module on SENet, used local cross-channel interaction strategy and the method of dynamically selecting the size of 1-D convolution kernel to enhance information interaction between adjacent channels, proved the importance of avoiding dimension reduction on channel attention, and showed good generalization ability in object detection and semantic segmentation tasks. CBAM [35] uses channel attention and spatial attention at the same time, uses the CAM module and the SAM module in turn to obtain effective features,

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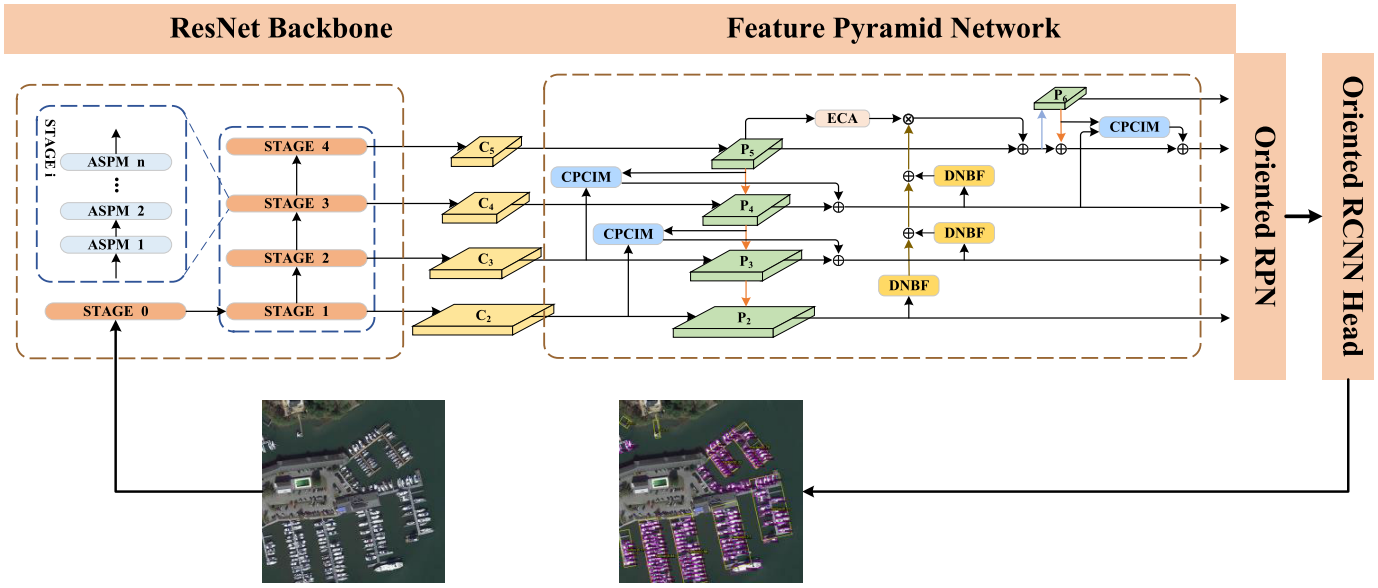


Fig. 2. Architecture of the proposed CPMFNet. CPMFNet mainly consists of four components: the backbone, FPN, oriented RPN, and oriented RCNN Head. By replacing the original resblock in the backbone network with our ASPM module, four different scale features are extracted. Five different scale-enhanced features are generated by feature fusion using FPN equipped with our proposed CPCIM module and DNBF filter. Oriented RPN and oriented RCNN Head use enhanced features to generate candidate anchor boxes, classification, and fine regression in turn.

and finally maps them into two different modules of channel and space to improve the feature map. SKNet [36] uses a selective kernel mechanism to dynamically select the size of the convolution kernel and uses an attention mechanism to adaptively adjust the receptive field, which effectively improves the performance of the convolutional neural network. ResNeSt [37] expands the idea of SKNet by dividing the input feature mapping into several groups. Li et al. [18] explored the large kernel selective mechanism in the field of remote sensing object detection for the first time. By designing a convolution module that can dynamically adjust the receptive field to build a backbone network LSKNet, which can more effectively deal with the feature extraction ability of different objects in complex backgrounds, and can dynamically obtain rich context information for different objects, combined with oriented R-CNN, SOTA performance is finally obtained on the DOTA-v1.0 dataset. Our method is the first to introduce a large kernel selective mechanism into FPN, which adaptively fuses across-scale feature information in spatial dimensions, and dynamically fuses high-level semantic features of deep network with precise positioning and detail features of shallow network.

III. Proposed Method

A. Oriented R-CNN as a Baseline

The detection speed of one-stage oriented object detection methods is fast, but there are many problems, such as the imbalance of positive and negative samples, and the difficulty of feature alignment. In the localization and detection of small objects, there are often missed detection, false detection, and other phenomena. The two-stage detection methods realize object detection through two independent networks, directly generate candidate regions on each pixel of the feature map, and then use a classifier to filter. Finally, the reserved candidate boxes continuously fit the real anchor box [38]. Therefore, we choose the competitive two-stage detection method oriented R-CNN [6] as our baseline model, use ResNet50-FPN as the backbone, and then use oriented RPN to generate high-quality oriented candidate boxes. Finally, the oriented RCNN head uses rotated RoIAlign to extract fixed feature vectors from the corresponding feature map for classification and fine regression of candidate boxes. Based on the baseline, we designed an adaptive spatial sensing module to replace the original residual block in the ResNet [39], inserted the constructed CPCIM module and DNBF filter into FPN [21], and finally proposed a new oriented object detection network called CPMFNet.

B. Overall Framework

The overall framework of the proposed CPMFNet is shown in Fig. 2. First, the image is input into ResNet. Due to the complex background of remote sensing images, the accurate detection of objects often depends on a wide range of context information. To effectively enhance dynamic context information for different target objects, we redesign the residual block in ResNet and propose ASPM, which dynamically adjusts the convolution kernel receptive field to adaptively mine context information and further enhance multiscale object features. The features extracted by the improved ResNet network are input into FPN. By inserting our CPCIM module and DNBF filter into FPN to fuse multiscale features, the problem of small object inundation is alleviated, and the object details and position features extracted by the shallow network are transferred to the deep network to make up for shortcomings of the deep network. Finally, the features generated by FPN are input into oriented RPN and oriented RCNN head in the baseline for subsequent candidate box generation, classification, and regression. In the neck network, the large kernel

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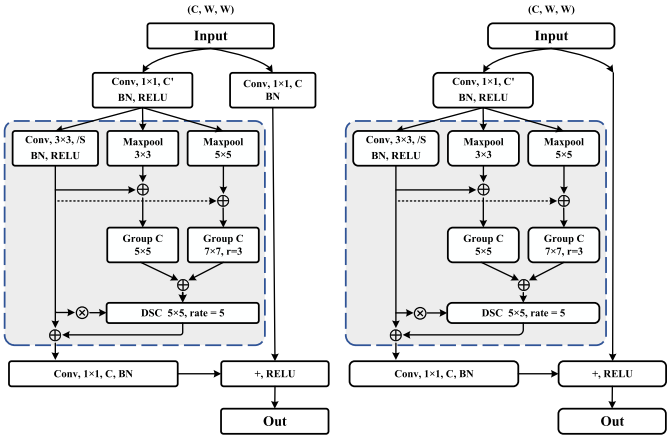


Fig. 3. Architecture of the proposed ASPM. Through the three-branch structure, the branch network of receptive fields with different sizes is realized to adaptively perceive spatial context information.

selective mechanism is used for cross-scale feature fusion. Benefiting from a large kernel selective mechanism, CPCIM aggregates rich deep global semantic features and fine shallow positional features to improve the performance of the model in detecting objects in complex backgrounds. DNBF is different from the bottom-up structure in PANet [27], abandoning the bottom-up network construction mode in sequence. Through different downsampling strides, the object details and position features are fused from bottom to top and finally guide the detection of small objects in the deepest network.

C. Adaptive Spatial Perception Module

There are large differences in object scales in remote sensing images, and different-scale objects require different receptive fields. Small receptive fields in a simple scene can extract context information required by the detected object. However, dynamic receptive fields are extremely needed in the complex background of remote sensing images. Dilated convolution [40] introduces a new parameter of dilation rate, which mainly defines the distance between values when the convolution kernel processes data and can provide a larger receptive field under the premise of an equal amount of calculation.

Group convolution [41] divides input feature maps into multiple groups and convolutes them in corresponding groups at the same time. Each group uses an independent convolution kernel for convolution calculation. Finally, all groups are combined to form the final output feature map. Compared with ordinary convolution, it reduces the number of parameters and introduces a certain degree of feature interaction. The depthwise separable convolution used in mobilenet [42] overlaps a point-by-point convolution after the group convolution to realize the function of group convolution channel fusion, which can reduce the amount of calculation while maintaining feature expression ability comparable to ordinary convolution.

We propose the ASPM module that combines dilated convolution, group convolution, and depthwise separable convolution. As shown in Fig. 3, the proposed ASPM module is based on the residual block in ResNet. The Bottleneck (BTNK) structure is introduced in a residual block of ResNet, which is divided into two structures with different input and output channels and the same number of input and output channels [39]. The ASPM module we proposed is also divided into two structures: the left side of Fig. 3 is the structure with different input and output channels, and the right side is the structure with the same number of input and output channels. Except for the difference in the number of input and output channels, the other two ASPM modules use the same structure. The boxed parts in the left and right parts of Fig. 3 represent the similarities between the two structures, that is, the core part of the ASPM module. Finally, the residual block with a BTNK-like structure in the ResNet50 backbone network is replaced by the proposed ASPM module. For the input feature, we first use 1×1 convolution for channel reduction to obtain $f \in R^{W \times H \times C}$, compared with traditional residual blocks, and then propose a three-branch architecture. The first branch is the traditional resblock branch, which can be represented as

$$f_{\text{res}} = \text{Conv}_{3\times3}(f, s_d = R_i) \tag{1}$$

among them, $\text{Conv}_{k\times k}(\cdot)$ represents a convolutional normalization activation operation with a kernel size of k, and $s_d = R_i$ represents downsampling stride is R_i, to obtain feature maps of the same shape, the value of the downsampling stride R_i will be adjusted according to the shape of input and output feature maps. The latter two branches, respectively, realize feature extraction of the larger receptive field convolution kernel. First, the feature f is downsampled by global max-pooling with a kernel size of 3 and 5, and then a 5×5 group convolution and a 7×7 group convolution with a dilation rate of 3 are used to realize feature extraction of different receptive field convolution kernel. The formula can be expressed as

$$b_1 = G_{5,1}\left[P_{\max}^{3\times3}(f, s_d = R_i) \oplus f_{\text{res}}\right] \tag{2}$$

$$b_2 = G_{7,3}\left[P_{\max}^{5\times5}(f, s_d = R_i) \oplus f_{\text{res}}\right] \tag{3}$$

where $G_{k,d}(\cdot)$ represents group convolution, $P_{\max}^{k\times k}$ represents global max-pooling, k and d are the kernel size and dilation rate, respectively, and $\oplus$ represents element-wise addition operation of feature maps.

The element-wise addition operation of feature maps can strengthen important features between two feature maps and reduce noise at the same time. The element-wise product operation of feature maps can enhance common features and weaken unimportant features, highlighting common features. The above dynamically receptive dual branches are fused via element-wise addition and then fed into a depthwise separable convolution module. This output is further element-wise produced with a traditional resblock branch to highlight common features. Finally, residual connections are added to obtain the three-branch enhanced feature f_e, which can be represented as

$$f_e = f_{\text{res}} \oplus \left[f_{\text{res}} \odot M_{(5,5)}(b_1 \oplus b_2)\right] \tag{4}$$

where $M_{(k,d)}$ represents a standard depthwise separable convolution block containing normalization and activation functions and $\odot$ represents element-wise product operation between feature maps.

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The ASPM uses a multibranch structure, and different branches correspond to different sizes of receptive field convolution layers. Finally, the object features extracted by adaptive receptive field convolution kernel corresponding to these branches are fused, and feature representation in scenes with large differences in object scales is improved by element-wise product operation with residual connection.

D. Feature Pyramid Network

As shown in Fig. 2, the backbone network serves as the foundation of the detection network to extract object features from remote sensing images and obtain feature maps $C_i \in R^{W\times H\times C}$, which are then transferred to FPN for multiscale feature fusion to obtain $P_i \in R^{W\times H\times C}$. The feature maps extracted by shallow networks have a larger size and smaller receptive field, so they can perceive more information such as object details and position information. Based on large feature maps extracted by shallow networks, multiscale feature maps are obtained through further feature extraction. The deeper the network, the smaller the feature maps and the superposition of receptive fields enables the network to learn more global semantic features of images. We take FPN as a reference model and combine the advantages of the dual-path feature fusion network proposed by PANet [27]. In the bottom-up stage, the proposed CPCIM takes shallow C_{i-1} and deep P_{i+1} as inputs and uses spatial kernel selective mechanism to effectively fuse cross-scale feature information from large convolution kernels of different scales. Finally, the fusion of details, positions, and other features of shallow networks with global semantic features of deep networks is achieved through element-wise addition operation. Compared with PANet, we have not fully adopted a bottom-up structure. Although shallow features have many fine detail features and high positioning accuracy, which can effectively assist deep networks and enhance semantic features, it is difficult to avoid a lot of noise in shallow networks. When fusing deep and shallow features, redundant noise will be introduced. Therefore, we propose a DNBF filter for dynamic pooling of P_2, P_3, and P_4 layers and finally jointly guide P_5 deep network to suppress shallow noise and assist in improving the detection ability of small objects in a deep network.

1) Composite Perception and Context Integration Module: The background of the remote sensing image is complex, and the detection of the oriented object needs more accurate object direction and position information to solve the problem of poor wrapping of generated bounding box to target object and improve detection accuracy. There are many noises in the shallow network, but due to high resolution and small receptive field, the shallow network is sensitive to the edge and position information of the target object. The deep network processes low-resolution feature maps, and while the network keeps deepening, it can obtain a broader global receptive field, which has rich semantic features and has strong anti-interference ability in the case of object deformation or a large number of background interference. By suppressing the noise of shallow networks and transferring more effective edge and position information to deep networks, the wrapping ability of bounding box-wrapping target objects can be further improved.

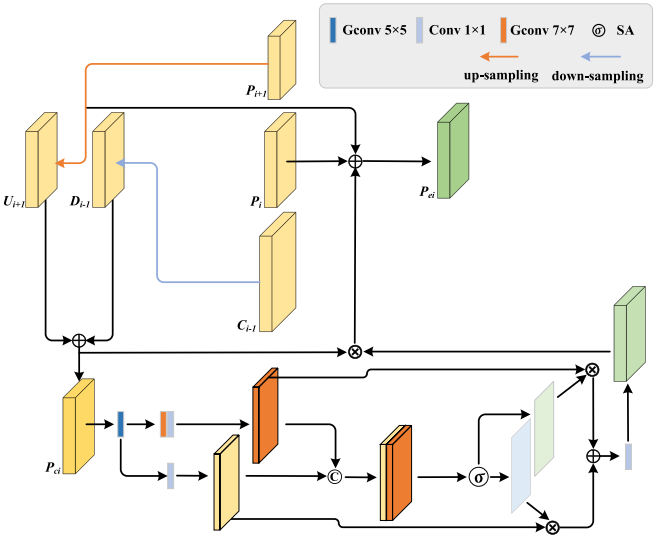


Fig. 4. Architecture of the proposed CPCIM.

Based on the above analysis, we insert CPCIM in the top-down stage, as shown in Fig. 4, C_{i-1} ($i \in \{3, 4, 5\}$) is downsampled to get D_{i-1}, and P_{i+1} is upsampled to get U_{i+1}, so that they have the same shape as P_i, and then D_{i-1} and P_{i+1} are added element by element for spatial fusion. It can be expressed as

$$P_{ci} = U(P_{i+1}) \oplus P_{\max}^{3\times 3}(C_{i-1}, s_d = 2) \tag{5}$$

where $U(\cdot)$ represents upsampling operation.

The wide and dynamic context information can optimize the detection ability of the model in complex backgrounds. The large kernel selective mechanism improves the dynamic range context extraction ability of the model through double-branch large kernel convolution and spatial attention mechanism, which can more effectively deal with background information of different ranges of objects. Finally, the shallow network and deep network are spatially merged, and edge and position features extracted by the shallow network are transferred to the deep network to guide the deep network to accurately locate objects and generate a more wrapping bounding box. It can be expressed as

$$b_1 = C_{1\times 1}\left[G_{5,1}(P_{ci})\right] \tag{6}$$

$$b_2 = C_{1\times 1}\left[G_{7,3}\left(G_{5,1}(P_{ci})\right)\right] \tag{7}$$

where $C_{1\times 1}(\cdot)$ represents a fully connected layer. The purpose of the sequential decomposition of large kernel sequences is to obtain a wider range of dynamic context information, reduce the amount of computation, and ensure that larger kernel learning than the first branch can be obtained. The spatial attention mechanism can well guide the model to focus on more important areas in the image. To better enhance interaction between different spaces, we first concatenate branches of different large kernel mechanisms and then use global average pooling and global max-pooling in channel dimension, respectively. The purpose of compressing the channel can greatly reduce the amount of calculation. Finally, the spatial attention graph $\text{Mask}_i \in [0, 1]^{W_{bi}\times H_{bi}\times 1}$ required by n different conv branches can be mapped through 1×1 convolution after

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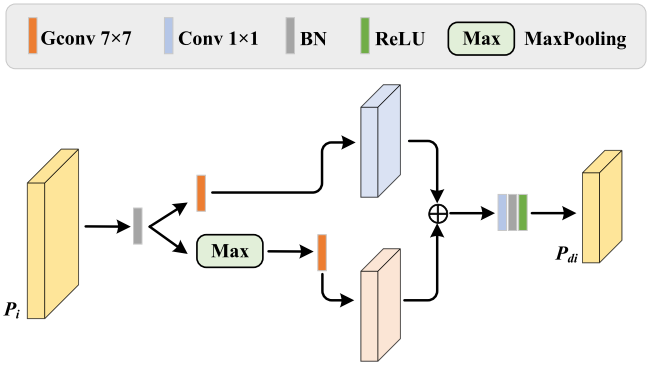


Fig. 5. Architecture of the proposed DNBF.

spatial interaction of different spatial features is strengthened by concat. The generation of spatial selection mask can be expressed as

$$\text{Mask}_i = \sigma_i[\text{Cat}(b_1, b_2)] \tag{8}$$

where σ_i stands for sigmoid function, and $\text{Cat}(\cdot)$ stands for feature map concatenation. Finally, the corresponding spatial attention map is used for weighted fusion, so that the fused features can have stronger semantic features and more edge position information at the same time. To enhance common features of the original P_i layer and fused features, we add the features of the C_i layer after the top-down stage and fused features element by element to improve the feature expression ability of the P_i layer. It can be expressed as

$$P_{ei} = P_i \oplus U_{i+1} \oplus C_{1\times1}\left[\sum_{i=1}^{2}(b_i \odot \text{Mask}_i)\right] \odot P_{ci}. \tag{9}$$

2) Detail-Noise Balance Filter: To suppress noise impact brought by the integration of shallow network and deep network, we abandoned direct element by element addition of P_i and P_{i-1} layers from bottom to top stage. Instead, we regarded P_2, P_3, and P_4 as the features of shallow network. The global semantic features concerned by the deepest network are the extreme lack of edge and position information of small objects. To better solve the problem of small object loss or low detection accuracy, we input all the above feature maps which are regarded as shallow network features into our proposed DNBF and finally get feature maps with the same shape as P_5 feature maps. As shown in Fig. 5, the DNBF first normalizes input. Since pooling has no learning parameters, the effect is not as good as the convolution downsampling method, but global max-pooling can obtain a wider range of spatial feature information without increasing parameters. Therefore, we add a 7×7 channel by channel convolution with a dilation rate of 3 after global max-pooling, combined with a direct large kernel channel by channel convolution downsampling method, it can be expressed as

$$F_i = G_{7,3}\left[P_{\max}^{9\times9}(P_i, s_d = R_i)\right] \oplus G_{7,1}(P_i, s_d = R_i). \tag{10}$$

Finally, the two branches are convoluted point by point, which can be expressed as

$$P_{di} = \text{Conv}_{1\times1}(F_i). \tag{11}$$

This network structure can be compared to a variant of depthwise separable convolution, except that we are a dual-branch channel-wise convolution with a dilation rate.

ECA is an improvement of SENet. To achieve more efficient local channel interaction, we use ECA attention on P_5 to obtain dependence between channels, thereby generating an enhanced P_5 layer feature map. First, element-wise addition operation is performed on outputs of all DNBF filters. Next, this result is an element-wise product with the enhanced P_5 layer feature map. Finally, the result is element-wise added to the original P_5 layer to further enhance the feature expression ability and edge learning capabilities of the deepest network for small objects. It can be expressed as

$$P_5 = P_5 \oplus \text{ECA}(P_5) \odot \left(\sum_{i=2}^{4} P_{di}\right). \tag{12}$$

IV. Experiments

To evaluate the effectiveness of our method, we conducted experiments on four benchmark remote sensing datasets. This section mainly describes the following four contents: 1) dataset description; 2) implementation details; 3) ablation experiments and analysis; and 4) comparison with state-of-the-art methods.

A. Datasets

1) *DOTA-v1.0 Dataset:* DOTA-v1.0 [43] is one of the large datasets used for object detection in remote sensing images. It contains 2806 remote sensing images in total, including 188 282 examples of 15 categories: plane (PL), baseball diamond (BD), bridge (BR), ground track field (GTF), small vehicle (SV), large vehicle (LV), ship (SH), tennis court (TC), basketball court (BC), storage tank (ST), soccer ball field (SBF), roundabout (RA), harbor (HA), swimming pool (SP), and helicopter (HC). Each image has objects of different scales, shapes, and directions, with resolutions ranging from 800 × 800 to 4000 × 4000 pixels.
2) *DOTA-v1.5 Dataset:* DOTA-v1.5 [43] uses the same images as DOTA-v1.0, but extremely small instances (less than 10 pixels) are also annotated. In addition, it is released with a new category, Container Crane (CC), and contains a total of 403 318 instances. The number of images and dataset division is the same as DOTA-v1.0.
3) *DIOR-R Dataset:* DIOR-R [44] is a massive object detection benchmark dataset for RSI. The dataset includes 23 463 optical images and 192 472 instances from 20 different object categories. These 20 object categories are airplane (APL), airport (APO), baseball field (BF), basketball court (BC), bridge (BR), chimney (CH), dam (DAM), expressway service area (ESA), expressway toll station (ETS), golf field (GF), ground track field (GTF), harbor (HA), overpass (OP), ship (SH), stadium (STA), storage tank (STO), tennis court (TC), train station (TS), vehicle (VE), and windmill (WM).
4) *HRSC2016 Dataset:* HRSC2016 [45] is a high-resolution remote sensing image dataset for ship detection. It consists of 1061 images, including 2976 ship

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TABLE I

ABLATION STUDIES OF DIFFERENT COMPONENTS ON THE DOTA-V1.0 TEST SET. A REPRESENTS ASPM, C REPRESENTS CPCIM, AND D REPRESENTS DNBF

Methods	ASPM	CPCIM	DNBF	mAP
Baseline				75.87
Baseline+A	✓			77.18
Baseline+C		✓		77.17
Baseline+D			✓	76.86
Baseline+A+C+D	✓	✓	✓	**78.23**

instances. Although the HRSC2016 dataset has a single ship category, the rotation object in the dataset is also difficult and challenging due to great changes in rotation, size, position, shape, and appearance of the HRSC2016 instance.

B. Implementation Details

For the sake of fairness, we adopt the same dataset processing method as other mainstream methods [6], [18], [20], [46]. Our model is built with oriented R-CNN as a baseline. All models are trained on the training set and validation set and finally tested on the test set, unless otherwise specified. For DOTA-v1.0 and DOTA-v1.5 datasets, the single-scale training strategy is adopted. We crop the original image into 1024 × 1024 patches in the form of a sliding window, with a stride size of 200, and use both horizontal and vertical flipping for data enhancement to prevent overfitting. We trained our model 14 epochs, tested it with a test set, and submitted the results on the DOTA dataset official website online server. For the DIOR-R dataset, we use the original image size of 800 × 800, and the training strategy is consistent with the DOTA-v1.0 dataset. For detection accuracy on the HRSC2016 dataset, we adjusted image size to 800 × 800 without changing the image scale, trained with 36 epochs, and used mean average precision (mAP) as evaluation metric, including PASCAL VOC2007 and VOC2012 metrics.

During the experiment, the AdamW optimizer was used for training, and weight decay was set to 0.05. Considering the problem of computational power in the first three datasets, we used two Tesla V100-SXM2-32 GB GPUs for model training, and for the last dataset, we used two RTX 2080Ti GPUs for model training. The initial learning rate of all datasets was 0.0001. All FLOPs reported in this article were calculated with a 1024 × 1024 image input, FPS on DOTA-v1.0, DOTA-v1.5, and HRSC2016 datasets were calculated with 1024 × 1024 image inputs, and the DIOR-R dataset was calculated with 800 × 800 image inputs.

C. Ablation Study

This section mainly reports the ablation experiment results of our model using a single-scale training strategy on the DOTA-v1.0 and DOTA-v1.5 datasets, and further studies the effectiveness of our method using oriented R-CNN of ResNet50-FPN as a baseline.

1) Overview: First, we conducted ablation experiments on the design of each part of the network. As shown in Table I, the detection accuracy of the model will be continuously improved by continuously adding proposed modules to the baseline. It not only proves the effectiveness of each component of our model but also proves the overall effectiveness of our method. In addition, we also made a visual analysis of our method and the baseline method. As shown in Fig. 7(a)–(e) show the visualization results of baseline, and (f)–(j) show visualization results of our model. By comparing and analyzing the detection effect of the red dotted circle area, we can find that our method can greatly reduce the phenomenon of missed detection and false detection of small objects. At the same time, our method can better locate the angle, position, and other information of the object, better wrap the object, and greatly improve the detection performance of baseline in some categories.

TABLE II

ABLATION STUDIES OF DIFFERENT COMPONENTS ON CC CLASS OF THE DOTA-V1.5 TEST SET. A REPRESENTS ASPM, C REPRESENTS CPCIM, AND D REPRESENTS DNBF

Methods	ASPM	CPCIM	DNBF	mAP
Baseline				30.61
Baseline+A	✓			31.86
Baseline+C		✓		32.83
Baseline+D			✓	32.23
Baseline+A+C+D	✓	✓	✓	**34.65**

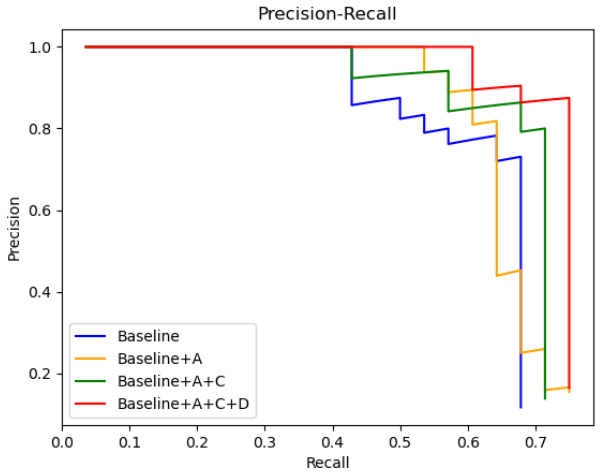


Fig. 6. Comparison of detection results between CPMFNet (Ours) and baseline on CC class of thevDOTA-v1.5 validation set.

To further evaluate the detection performance of the proposed method for small objects, we conducted ablation experiments on the DOTA-v1.5 dataset. We selected the newly added CC category in the DOTA-v1.5 dataset for ablation experiments. As shown in Table II, by adding our proposed modules to the baseline, respectively, the detection accuracy of the CC category is improved. After inserting all the proposed modules into the baseline, the detection result of the CC category is improved by 4.04% compared with the baseline. To more intuitively compare the performance difference between the proposed method and the baseline, Fig. 6 shows the PR curves comparison of the baseline model and the model with proposed modules inserted on the CC category of DOTA-v1.5 validation set. By comparing the red PR curve

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TABLE III
INFLUENCE OF HYPERPARAMETERS IN ASPM ON THE DOTA-V1.0 TEST SET

(k1, d1)	(k2, d2)	DSC	RF1	RF2	mAP
(3, 1)	(5, 2)	(5, 5)	23	28	77.27
(5, 1)	(7, 3)	(5, 5)	25	38	**78.23**
(7, 1)	(9, 4)	(5, 5)	27	52	77.57
(5, 1)	(7, 3)	(5, 3)	17	30	77.77
(5, 1)	(7, 3)	(5, 7)	33	46	77.63

TABLE IV
INFLUENCE OF USING CPCIM IN DIFFERENT LAYERS ON THE DOTA-V1.0 TEST SET

Module	P_3	P_4	P_5	mAP
CPCIM	✓			77.09
CPCIM	✓	✓		77.81
CPCIM		✓	✓	77.59
CPCIM	✓	✓	✓	**78.23**

TABLE V
INFLUENCE OF HYPERPARAMETERS IN DNBF ON THE DOTA-V1.0 TEST SET

Pooling	GroupC	mAP
Max.	(5, 3)	77.23
Avg.	(5, 3)	77.04
Max.	(7, 3)	**78.23**
Avg.	(7, 3)	77.52

TABLE VI
INFLUENCE OF USING THE DNBF MODULE IN DIFFERENT LAYERS ON THE DOTA-V1.0 TEST SET

Module	P_2	P_3	P_4	mAP
DNBF	✓			77.19
DNBF	✓	✓		77.36
DNBF		✓	✓	77.63
DNBF	✓	✓	✓	**78.23**

of the proposed method in Fig. 6 with the blue PR curve of baseline, the effectiveness of the proposed method for small object detection is further verified.

2) Effectiveness of ASPM: For ASPM, it needs to fuse features of different receptive field sizes, so the receptive field of a convolutional kernel of a branch network is particularly critical. We configure ASPM as shown in Table III. The receptive field is defined as

$$\mathrm{RF}_1 = k_1, \mathrm{RF}_i = d_i(k_i - 1) + \mathrm{RF}_{i-1}. \tag{13}$$

When $(k1, d1)$, $(k2, d2)$, and DSC, respectively, take $(5, 1)$, $(7, 3)$, and $(5, 5)$, the model detection performance is the best, which also indicates that we can improve model detection performance by selecting convolutional branches with fused receptive fields of 25 and 38. Too large or too small receptive fields of branches will affect the performance of our module.

3) Effectiveness of CPCIM: The composite perception and context integration module (CPCIM) is a key module that integrates higher-level global semantic representation information of deep layer network and edge, location features of the shallow layer network, and suppresses redundant noise in a shallow layer. As shown in Table IV, we can find that adding CPCIM in part of P_3, P_4, and P_5 layers is not as effective as adding them all. The reason is that with the deepening of the network, the information on small objects is gradually lost. In a deep network, we need some precise positioning information in the shallow network, more specifically, the feature information of some edges or details of the object in remote sensing image in the initial learning stage of the network. The results shown in Table IV also show that the proposed CPCIM can not only suppress shallow noise but also transmit more small object information to the deep network, which improves detection effect of the model.

4) Effectiveness of DNBF: The DNBF needs to transmit all features that we regard as shallow networks to the deepest layer of the network while suppressing noise. It can also be understood as the network layer that is the most lacking in small object information. Therefore, it is necessary to set corresponding downsampling strides for different layers. If the downsampling stride is too large, it is easy to lose a large amount of spatial feature information. For the selection of the max-pooling kernel, the stride required to sample the shallowest feature down to the deepest network is 8, when the pooling stride is greater than kernel size, the loss will be NaN through the experiment, so the default kernel of max-pooling is set to 9. As shown in Table V, we performed ablation experiments on the downsampling method. The kernel size and dilation rate in Group C represent both the kernel size and dilation rate of group convolution after pooling in the first branch and the size of the convolution downsampling kernel in the second branch. As the results in Table V, the global max-pooling can retain more local information of shallow network, especially the texture features, which is more suitable for capturing edges and details in the image. The effect is best when the group convolution kernel is 7. Although a large convolution kernel is not particularly popular in deep learning, it involves direct downsampling from the shallowest layer to the deepest layer. Therefore, if the size of the downsampling step is smaller than the convolution kernel for a long time, some edge information will be lost across some pixels. We also set up ablation experiments similar to the CPCIM module. As shown in Table VI, we set up to transfer features of different network depths to a deep network that needs the assistance of the shallow network most. The final results also show that we can provide the best detection performance by taking P_2, P_3, and P_4 together as the way of the shallow network to jointly guide the deepest network.

5) Parameters and Computational Cost of Network: As shown in Table VII, the parameters and computational cost of the proposed method and baseline are listed. The input scale of 1024×1024 pixels is used for FLOPs calculation. Compared with the baseline, the proposed method has only a slight increase in parameters and FLOPs, but the detection

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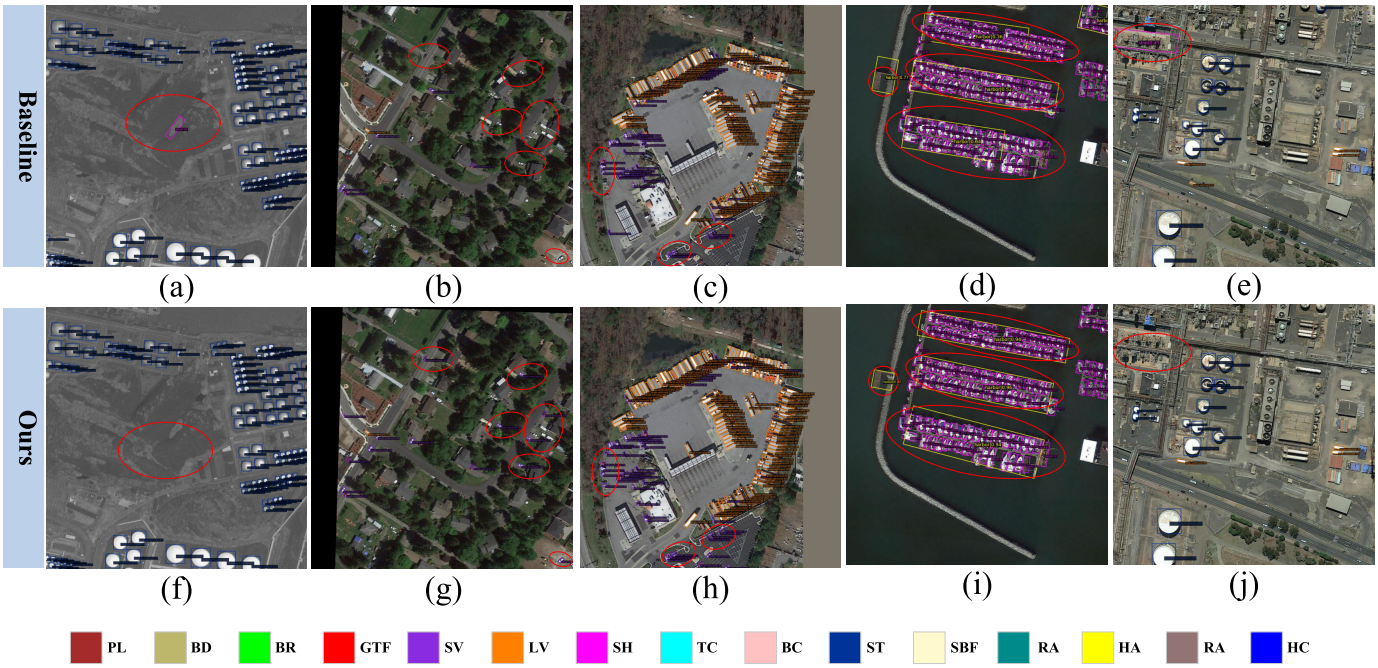


Fig. 7. Comparison of detection results between CPMFNet (ours) and the baseline. (a)–(e) Show the visualization results of baseline and (f)–(j) show visualization results of our model.

TABLE VII
COMPLEXITY, HYPERPARAMETERS, AND ACCURACY COMPARISON

	Datasets	Baseline	CPMFNet (Ours)
#P	All	41.1M	42.3M
FLOPs	All	211G	218G
mAP (%)	DOTA-v1.0	75.87	78.23 **(+2.36)**
	DOTA-v1.5	67.60	70.85 **(+3.25)**
	DIOR-R	64.30	67.58 **(+3.28)**
	HRSC2016	96.50	98.61 **(+2.11)**

accuracy on four datasets has been significantly improved. It can be seen that the proposed method has improved by 2.36% compared with the baseline on the DOTA-v1.0 dataset. To evaluate the performance of small object detection, we also evaluated the DOTA-v1.5 dataset, which is finally 3.25% better than the baseline. On the DIOR-R and HRSC2016 datasets, it also increased by 3.28% and 2.11% compared with the baseline, respectively. These further verify the effectiveness and feasibility of the proposed method.

D. Comparison With State-of-the-Art Methods

1) Results and Analysis on DOTA-v1.0 Dataset: We evaluated our method on the DOTA-v1.0 dataset and compared it with the 22 most advanced methods at present. We divided the comparison methods into anchor-free methods, one-stage anchor-based methods, and two-stage anchor-based methods. Our method belongs to the two-stage anchor-based method. As shown in Table VIII, the proposed method achieves SOTA detection performance under a single-scale training strategy with an mAP of 78.23%, and a gain of 2.36% compared with the baseline [6]. In anchor-free methods, we achieved a 2.26% mAP improvement compared to using the same R-50-FPN backbone with Oriented RepPoints. In addition, compared to using Swin-T-FPN with Oriented RepPoints, we also saw a 0.6% mAP increase. When compared to one-stage anchor-based methods, our method significantly outperforms many methods. Specifically, compared to AO2-DETR based on Transformers, our CNN-based method shows a 0.5% improvement. In comparison with two-stage anchor-based methods, our method achieves SOTA performance. Only when ARC uses the ARC-R101-FPN backbone can the performance gap between ARC and ours using R-50-FPN be narrowed to around 0.5% mAP. In terms of inference speed, it can be seen from Table VIII that some one-stage detection methods have lower FLOPs but slower inference speeds. Although our model is slightly higher than the baseline model in FLOPs, we improve the parallel processing capability of the model by extensively using optimization strategies such as element-wise operations, group convolutions, and depth-wise separable convolutions in network module design, thereby achieving faster inference speed. Finally, our model achieves a faster inference speed than baseline and ranks first among all compared methods with an inference speed of 18.6 frames/s. Although our computational complexity is 218G, compared with methods with the same computational complexity or even some methods with higher computational complexity, our method achieves higher detection accuracy and achieves a better tradeoff between detection speed, detection accuracy, and model complexity. In addition to quantitative analysis, we also made a visual analysis of our model on the DOTA-v1.0 dataset, as shown in Fig. 8.

2) Results and Analysis on the DOTA-v1.5 Dataset: The DOTA-v1.5 dataset annotates very small instances (less than 10 pixels) based on the DOTA-v1.0 dataset. To better evaluate detection performance in small object scenarios, we evaluate

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TABLE VIII

COMPARISON WITH STATE-OF-THE-ART METHODS ON THE DOTA-V1.0 DATASET. RED AND BLUE REPRESENT THE BEST AND THE SECOND-BEST PERFORMANCE, RESPECTIVELY

	Methods	Backbone	PL	BD	BR	GTF	SV	LV	SH	TC	BC	ST	SBF	RA	HA	SP	HC	mAP	FLOPs	FPS
Anchor-free	CFA [47]	R-50-FPN	88.04	82.14	53.90	73.69	79.94	78.87	87.16	90.87	81.90	85.63	56.14	64.40	70.31	70.63	38.05	73.45	194G	16.1
	CFA [47]	R-101-FPN	89.26	81.72	51.81	67.17	79.99	78.25	84.46	90.77	83.40	85.54	54.86	67.78	73.04	70.24	64.96	75.05	-	14.9
	G-Rep [9]	R-50-FPN	87.76	81.29	52.64	70.53	80.34	80.56	87.47	90.74	82.91	85.01	61.48	68.51	67.53	73.02	63.54	75.56	-	8.6
	Oriented-RepPoints [8]	R-50-FPN	87.02	83.17	54.13	71.16	80.18	78.40	87.28	90.90	85.97	86.25	59.90	70.49	73.53	72.27	58.97	75.97	194G	16.1
	Oriented-RepPoints [8]	Swin-T-FPN	89.11	82.32	56.71	74.95	80.70	83.73	87.67	90.81	87.11	85.85	63.60	68.60	75.95	73.54	63.76	77.63	-	14.9
One-stage	R3Det [1]	R-50-FPN	89.29	75.21	45.41	69.24	75.54	72.89	79.29	90.89	81.02	83.25	58.81	63.15	63.43	62.21	37.41	69.80	336G	13.5
	DAL [48]	R-50-FPN	88.68	76.55	45.08	66.80	67.00	76.76	76.94	90.84	79.54	78.45	57.71	62.27	69.05	73.14	60.11	71.44	-	14.9
	R3Det-KLD [49]	R-50-FPN	89.20	75.61	48.33	73.02	76.87	75.29	86.35	90.90	84.52	83.46	60.83	62.10	66.56	64.90	43.86	72.12	336G	13.5
	S^2ANet [50]	R-50-FPN	89.11	82.84	48.37	71.11	78.11	78.39	87.25	90.83	84.90	85.64	60.36	62.60	65.26	69.13	57.94	74.12	198G	14.3
	SASM [7]	R-50-FPN	86.42	78.97	52.47	69.84	77.30	75.99	86.72	90.89	82.63	85.66	60.13	68.25	73.98	72.22	62.37	74.92	194G	15.3
	AO2-DETR [2]	R-50	89.27	84.97	56.67	74.89	78.87	82.73	87.35	90.50	84.68	85.41	61.97	69.96	74.68	72.39	71.62	77.73	304G	-
Two-stage	Faster R-CNN-O [26]	R-50-FPN	88.44	73.06	44.86	59.09	73.25	71.49	77.11	90.84	78.94	83.90	48.59	62.95	62.18	64.91	56.18	69.05	211G	14.1
	RoI Transformer [5]	R-101-FPN	88.64	78.52	43.44	75.92	68.81	73.68	83.59	90.74	77.27	81.46	58.39	53.54	62.83	58.93	47.67	69.56	225G	14.4
	AOPG [44]	R-50-FPN	89.27	83.49	52.50	69.97	73.51	82.31	87.95	90.89	87.64	84.71	60.01	66.12	74.19	68.30	57.80	75.24	-	12.4
	DODet [51]	R-50-FPN	89.34	84.31	51.39	71.04	79.04	82.86	88.15	90.90	86.88	84.91	62.69	67.63	75.47	72.22	45.54	75.49	-	14.9
	Oriented R-CNN [6]	R-50-FPN	89.46	82.12	54.78	70.86	78.93	83.00	88.20	90.90	87.50	84.68	63.97	67.69	74.94	68.84	52.28	75.87	211G	16.2
	Oriented R-CNN [6]	R-101-FPN	88.86	83.48	55.27	76.92	74.27	82.10	87.52	90.90	85.56	85.33	65.51	66.82	74.36	70.15	57.28	76.28	-	16.2
	AOPG+Smooth GIoU Loss [16]	R-50-FPN	89.62	82.89	52.24	72.34	78.27	83.08	88.18	90.89	87.10	84.48	63.49	66.55	72.64	59.10	62.03	76.19	-	-
	ReDet [46]	Re50-ReFPN	88.79	82.64	53.97	74.00	78.13	84.06	88.04	90.89	87.78	85.75	61.76	60.39	75.96	68.07	63.59	76.25	80G	10.9
	ARC [20]	ARC-R50-FPN	89.40	82.48	55.33	73.88	79.37	84.05	88.06	90.90	86.44	84.83	63.63	70.32	74.29	71.91	65.43	77.35	211G	-
	RF-Next-Oriented R-CNN [19]	RF-R-50-FPN	89.48	83.35	55.74	76.83	79.44	84.78	88.23	90.88	86.96	85.09	67.18	68.38	76.60	72.13	62.81	77.65	-	-
	ARC [20]	ARC-R101-FPN	89.39	83.58	57.51	75.94	78.75	83.58	88.08	90.90	85.93	85.38	64.03	68.65	75.59	72.03	65.68	77.70	290G	-
Ours	CPMFNet	R-50-FPN	89.53	82.92	54.63	77.58	79.37	84.87	88.23	90.95	86.95	84.93	68.2	70.39	76.35	72.01	66.68	78.23	218G	18.6

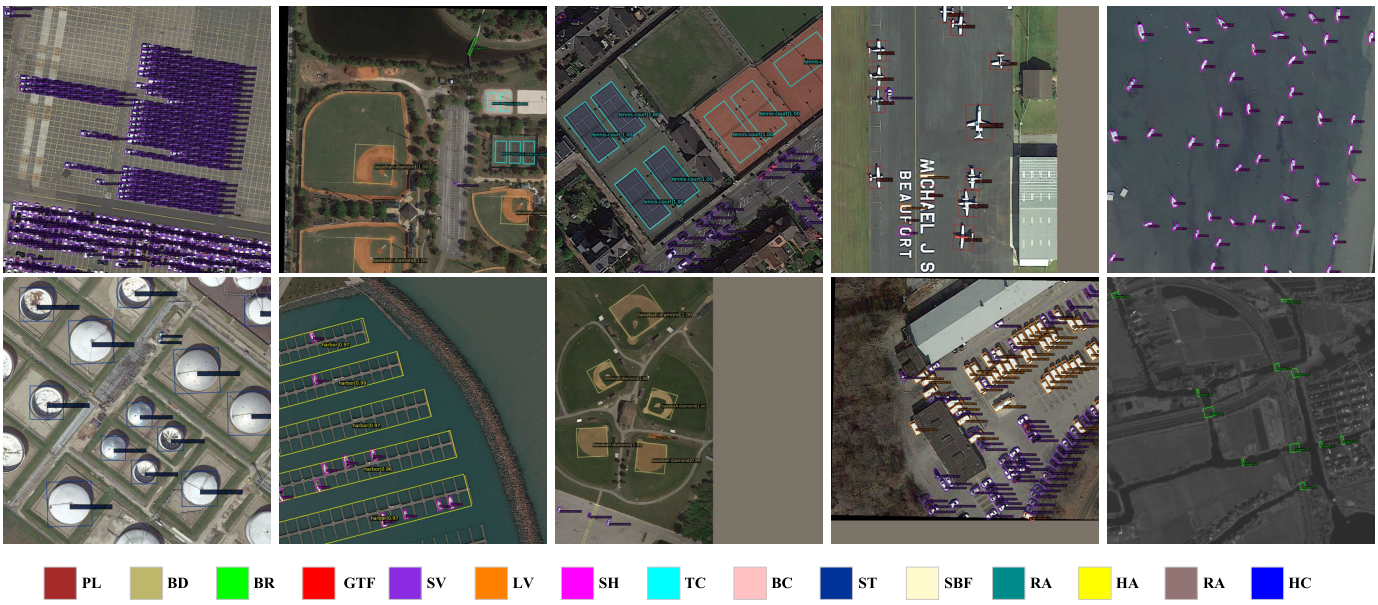


Fig. 8. Detection results of CPMFNet on the DOTA-v1.0.

the proposed method on the DOTA-v1.5 dataset. As shown in Table IX, we compare the model complexity, speed, and accuracy of different methods. Compared with 14 SOTA methods, our method achieves SOTA detection accuracy at a faster inference speed. The proposed method achieves a higher mAP detection accuracy of 70.85% with a comparable detection speed of 16.8 frames/s, ranks first in AP detection accuracy of GTF, LV, BC, SBF, HA, and CC classes, and ranks second in BR, SV, SH, TC, and RA classes. Compared with RetinaNet-FPNFormer [58], which ranks second in overall mAP detection accuracy, the mAP detection accuracy of our proposed method is improved by 2.62%, and 3.25% compared with the baseline [6].

3) Results and Analysis on the DIOR-R Dataset: DIOR-R is a more challenging remote sensing image-oriented object detection dataset. Our method has been evaluated and compared with some SOTA methods in terms of detection accuracy, model complexity, and speed. As shown in Table X, it represents the AP of each category of different methods on the DIOR-R dataset and mAP under comprehensive consideration of all categories. Finally, the proposed method obtains a comparable detection speed of 24.3 frames/s and achieves SOTA performance with 67.58% mAP, which gains 3.28% compared with the baseline. By analyzing the AP value of each category, our method achieves the best performance in APO, BC, BR, CH, DAM, GTF, STA, and TC categories, and the second-best detection accuracy in APL, GF, OP, and TS categories. Especially in the CH, APO, STA, and DAM categories, it surpassed the second-ranking method (mAP) by 3.88%, 2.84%, 1.58%, and 1.21%, respectively. Finally, some visualization results of our method on the DIOR-R dataset are shown in Fig. 9.

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TABLE IX

COMPARISON WITH STATE-OF-THE-ART METHODS ON THE DOTA-V1.5 DATASET. RED AND BLUE REPRESENT THE BEST AND THE SECOND-BEST PERFORMANCE, RESPECTIVELY

Methods	Backbone	PL	BD	BR	GTF	SV	LV	SH	TC	BC	ST	SBF	RA	HA	SP	HC	CC	mAP	FLOPs	FPS
RetinaNet-O [31]	R-50-FPN	71.43	77.64	42.12	64.65	44.53	56.79	73.31	90.84	76.02	59.96	46.95	69.24	59.65	64.52	48.06	0.83	59.16	203G	12.1
EMO2-DETR [52]	R-50-FPN	71.81	75.36	45.09	58.70	48.19	73.26	80.28	90.70	73.05	76.53	39.36	65.31	59.96	69.29	47.11	15.64	61.67	-	-
Faster R-CNN-O [26]	R-50-FPN	71.89	77.64	44.45	59.87	51.28	68.98	79.37	90.78	77.38	67.50	47.75	69.72	61.22	65.28	60.47	1.54	62.00	211G	14.1
Mask R-CNN-O [53]	R-50-FPN	76.84	73.51	49.90	57.80	51.31	71.34	79.75	90.46	74.21	66.07	46.21	70.61	63.07	64.46	57.81	9.42	62.67	-	-
HTC-O [54]	R-50-FPN	77.80	73.67	51.40	63.99	51.54	73.31	80.31	90.48	75.12	67.34	48.51	70.63	64.84	64.48	55.87	5.15	63.40	-	7.9
AOPG [44]	R-50-FPN	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	64.41	-	-
AO2-DETR [2]	R-50-FPN	79.55	78.14	42.41	61.23	55.34	74.50	79.57	90.64	74.76	77.58	53.56	66.91	58.56	73.11	69.64	24.71	66.26	304G	-
GGHL [55]	DarkNet-53	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	66.48	-	-
Oriented Rep [8]	R-50-FPN	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	66.71	194G	-
DCFL [56]	Re50-ReFPN	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	66.80	-	-
ReDet [46]	Re50-ReFPN	79.20	82.81	51.92	71.41	52.38	75.73	80.92	90.83	75.81	68.64	49.29	72.03	73.36	70.55	63.33	11.53	66.86	80G	9.9
Oriented R-CNN [6]	R-50-FPN	77.07	79.24	50.12	70.56	50.07	74.83	85.37	88.32	78.83	66.09	59.95	70.85	71.36	69.89	59.12	30.61	67.60	211G	-
RiDOP [57]	Re50-ReFPN	79.98	83.09	55.21	65.82	52.39	76.34	87.99	90.90	81.52	67.34	49.88	70.85	73.09	73.08	68.54	11.79	67.99	-	10.4
RetinaNet-FPNFormer [58]	R-50-FPNFormer	80.97	84.46	49.25	72.44	50.12	73.37	86.10	90.42	80.54	73.49	60.41	74.09	63.11	72.90	65.64	14.30	68.23	263G	-
CPMFNet (Ours)	R-50-FPN	79.25	81.68	54.76	73.52	54.20	77.23	87.51	90.88	83.50	68.22	63.50	73.69	74.56	72.55	64.30	34.57	70.85	218G	16.8

TABLE X

COMPARISON WITH STATE-OF-THE-ART METHODS ON THE DIOR-R DATASET. RED AND BLUE REPRESENT THE BEST AND THE SECOND-BEST PERFORMANCE, RESPECTIVELY

Methods	APL	APO	BF	BC	BR	CH	DAM	ETS	ESA	GF	GTF	HA	OP	SH	STA	STO	TC	TS	VE	WM	mAP	FLOPs	FPS
Faster RCNN-O [26]	62.79	26.80	71.72	80.91	34.20	72.57	18.95	66.45	65.75	66.63	79.24	34.95	48.79	81.14	64.34	71.21	81.44	47.31	50.46	65.21	59.54	211G	19.0
Gliding Vertex [25]	65.35	28.87	74.96	81.33	33.88	74.31	19.58	70.72	64.70	72.30	78.68	37.22	49.64	80.22	69.26	61.13	81.49	44.76	47.71	65.04	60.06	198G	15.2
ASDet [59]	62.88	30.33	71.41	81.43	36.84	72.51	25.08	74.94	60.65	66.87	80.62	41.92	52.88	81.21	74.62	71.00	81.48	54.28	47.49	62.91	61.57	-	24.5
RoI Transformer [5]	63.34	37.88	71.78	87.53	40.68	72.60	26.86	78.71	68.09	68.96	82.74	47.71	55.61	81.21	78.23	70.26	81.61	54.86	43.27	65.52	63.87	225G	13.4
Oriented R-CNN [6]	71.10	39.30	79.50	86.20	43.10	72.60	29.50	66.70	79.30	68.60	82.40	43.60	57.60	81.30	74.20	62.60	81.40	54.80	46.80	66.00	64.30	211G	21.2
AOPG [44]	62.39	37.79	71.62	87.63	40.90	72.47	31.08	65.42	77.99	73.20	81.94	42.32	54.45	81.17	72.69	71.31	81.49	60.04	52.38	69.99	64.41	-	-
DODet [51]	63.40	43.35	72.11	81.32	43.12	72.59	33.32	78.77	70.84	74.15	75.47	48.00	59.31	85.41	74.04	71.56	81.52	55.47	51.86	66.40	65.10	-	-
CGHL [55]																					66.48	-	-
AOPG+Smooth GIoU [16]	69.87	40.66	77.28	87.77	41.13	72.64	32.07	63.65	74.49	73.57	81.73	46.91	54.58	89.07	71.68	77.31	88.45	59.23	52.99	70.15	66.51	-	-
Oriented RepPoints [8]	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	66.71	194G	-
PIIDet-101 [60]	77.90	31.84	79.37	89.99	44.96	72.72	30.74	79.16	80.06	76.59	81.28	44.26	57.94	89.04	69.28	73.10	59.50	56.96	59.87	62.39	67.35	-	23.1
CPMFNet (Ours)	71.75	46.19	78.81	90.03	45.01	78.19	34.53	69.27	78.83	76.58	83.29	45.59	58.91	81.09	79.81	69.88	88.98	59.42	48.58	65.91	67.58	218G	24.3

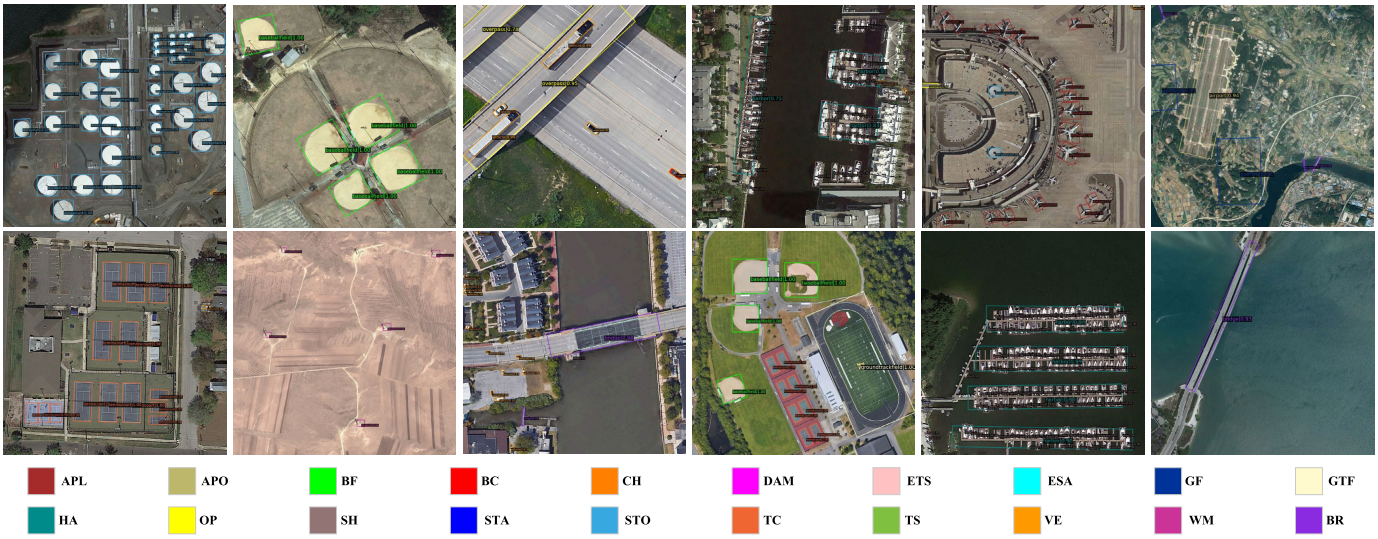


Fig. 9. Detection results of CPMFNet on the DIOR-R.

4) Results and Analysis on the HRSC2016 Dataset: Most ships in the HRSC2016 dataset are densely distributed. To evaluate the performance of the proposed method for object detection in scenes with dense distribution and high coincidence, the detection performance of our method on this dataset is quantitatively analyzed with 13 currently advanced methods. Table XI shows the comparison results of all methods in terms of detection accuracy, model complexity, and speed. Using the PASCAL VOC2012 metric, our model surpasses all current SOTA methods and achieves the best performance on the current dataset as we know. Using the PASCAL VOC2007 metric, our method ranks third after STD-ViT-B and LSKNet, but the difference is very small. Compared with the Transformer model of STD-ViT-B, our CNN model can surpass them in the PASCAL VOC2012 metric, which proves the effectiveness of our method. In terms of inference speed, we achieved an inference speed of 16.2 frames/s, second only to SASM [7]. We also found that the inference speed of our

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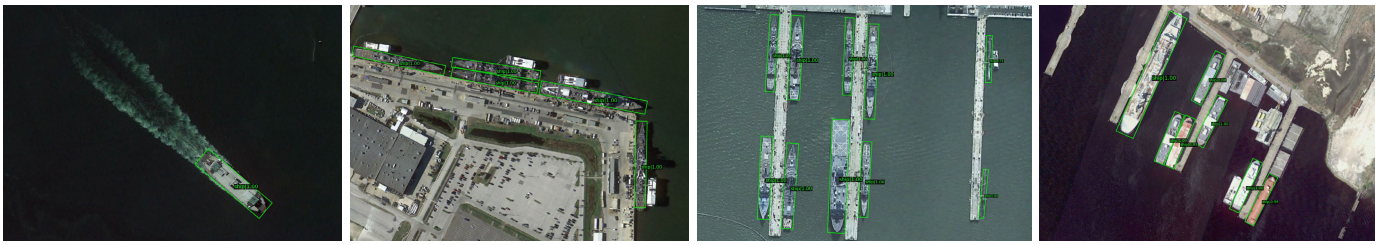


Fig. 10. Detection results of CPMFNet on the HRSC2016.

TABLE XI
COMPARISON WITH STATE-OF-THE-ART METHODS ON THE HRSC2016 DATASET

Methods	mAP(07)	mAP(12)	FLOPs	FPS
RoI Transformer [5]	86.20	-	225G	14.0
AO2-DETR [2]	88.12	97.47	304G	-
SASM [7]	90.00	-	194G	18.2
S^{2}ANet [50]	90.17	95.01	198G	16.0
PSD-SQ [61]	90.23	98.12	-	-
AOPG [44]	90.34	96.22	-	12.4
Oriented RepPoints [8]	90.38	97.26	194G	-
Oriented R-CNN [6]	90.40	96.50	211G	15.1
ReDet [46]	90.46	97.63	80G	13.2
RiDOP [57]	90.54	98.47	-	-
RTMDet [62]	90.60	97.10	216G	-
LSKNet-S [18]	90.65	98.46	172G	-
STD-ViT-B [63]	**90.67**	98.55	-	-
CPMFNet (Ours)	90.62	**98.61**	218G	16.2

model on multicategory complex dataset DOTA-v1.0 is better than that of SASM [7]. We analyzed the difference in inference speed caused by resolution and data distribution. The reason is that the detection speed of the model in a single-category simple scene dataset will be affected by some complex network structures. These complex network structures are specially designed for object detection in complex scenes, which ultimately requires more calculation and processing during the inference process. However, in terms of detection accuracy, our proposed method has a considerable improvement over SASM [7] under the PASCAL VOC2007 metric, which is sufficient to prove that the proposed method achieves a better tradeoff between detection accuracy and detection speed and has a certain superiority. The part of visualization results of our method on the HRSC2016 dataset are shown in Fig. 10. Except for STD-ViT-B and LSKNet methods, the rest of the methods use ResNet50-FPN as the backbone network for evaluation.

V. CONCLUSION

In this article, we propose a new oriented object detection network CPMFNet. First, aiming at the problems of large object scale differences and incomplete context information mining in remote sensing images, we propose ASPM to dynamically adjust the feature extraction receptive field, strengthen spatial perception, and enhance object feature representation. Second, the FPN is reconstructed. Specifically, we design CPCIM, which introduces a large kernel selective mechanism in the neck network to integrate global semantic features in deep network and edge, detail features in shallow network. We propose DNBF to dynamically suppress shallow network noise, enhance features of small objects in deep networks, and greatly reduce problems of missing detection and false detection of small objects. Finally, a large number of experiments on three benchmark datasets (DOTA-v1.0, DIOR-R, and HRSC2016) also prove the SOTA performance of our method in remote sensing image object detection.

In future work, we will focus on how to further design higher quality OBB representations based on this article, especially how to generate more accurate anchor box representations for large aspect ratio target objects, improve the accuracy of large-aspect-ratio target detection, and break through the bottleneck of overall class detection accuracy of AOOD tasks. In addition, we will further study how to reduce the impact of resolution and data distribution differences on model inference speed in future research work.

Author Contributions: Peng Bai: Conceptualization, methodology, and write an original draft. Ying Xia: Supervision, writing—review and editing. Jiangfan Feng: Investigation and funding acquisition.

DECLARATIONS

Data Availability: The datasets used during the study are available at https://captain-whu.github.io/DOTA/dataset.html, https://aistudio.baidu.com/datasetdetail/251054, and https://www.kaggle.com/datasets/guofeng/hrsc2016.

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